

SMARTX: Porewater Methane

May 2026 Samples

2026-07-17

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##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

Run Information: Brenden D. Blakley

```
#set the run date & user name
sample_year <- "2026"
sample_month <- "May"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_files <- c("Raw Data/20260625_SMARTX_GHG_May2026.csv", "Raw Data/20260629_SMARTX_GHG_May2026.csv",

# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/SMARTX_ShimadzuGC_QAQC_2026.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_SMARTX_Porewater_CH4_202605.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- ""

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

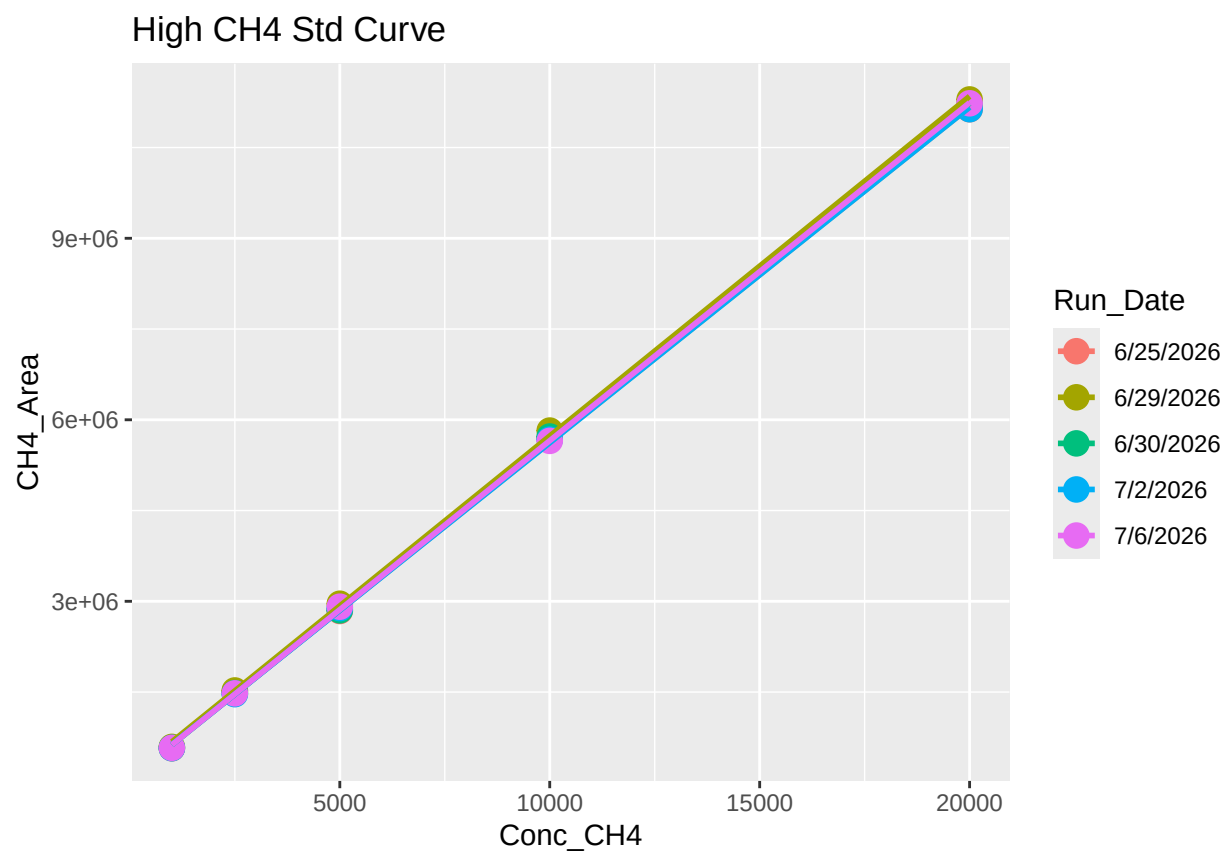
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

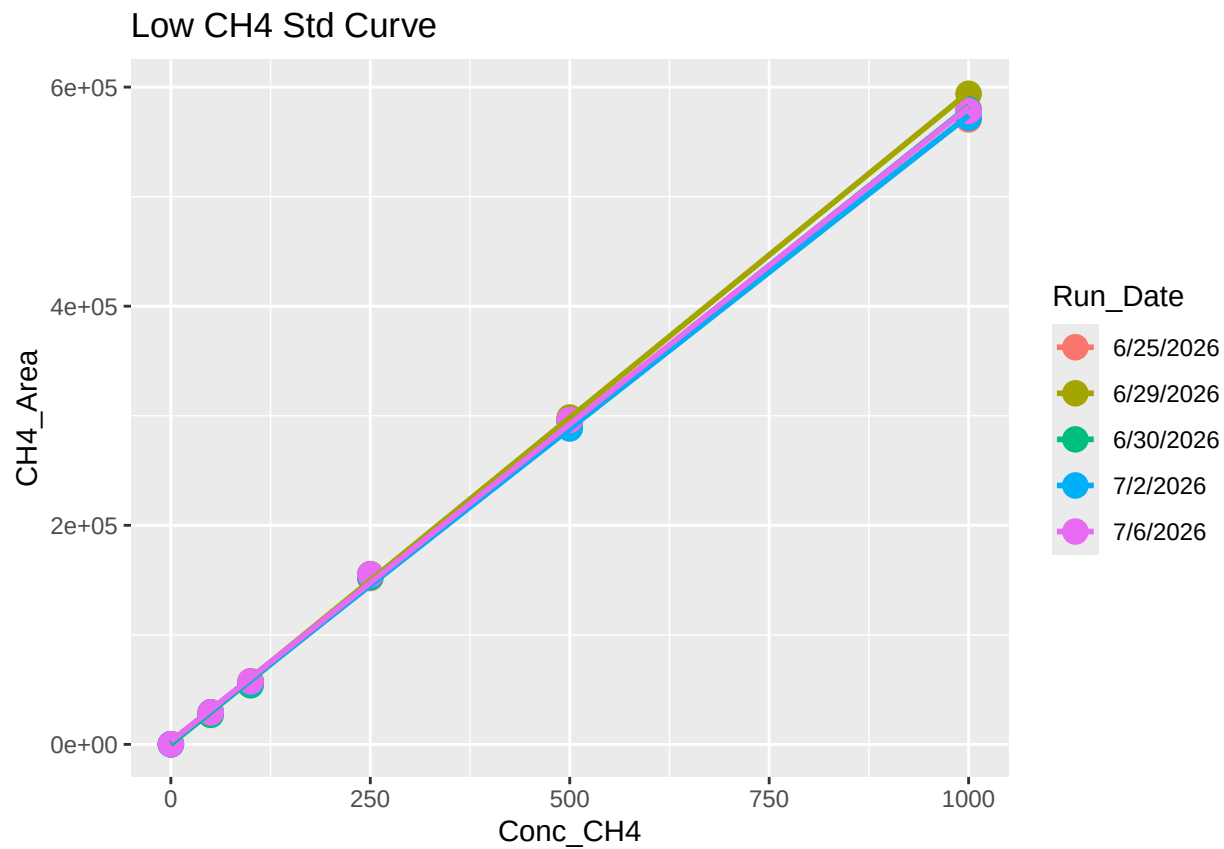
1 Read in data files

2 Assess standard curves

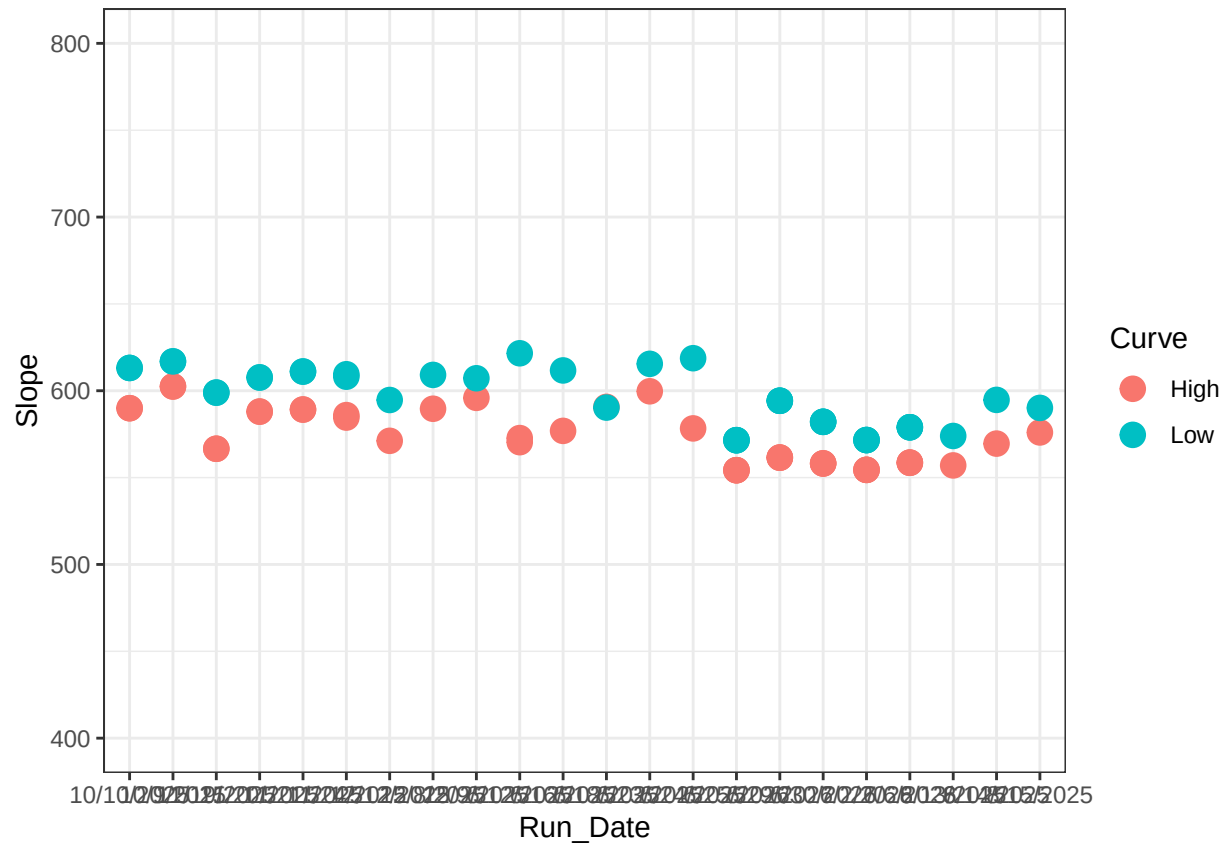
'geom_smooth()' using formula = 'y ~ x'



```
## 'geom_smooth()' using formula = 'y ~ x'
```



##	X	Curve	R2	Slope	Intercept	Run_Date
## 1	1	High	0.9986295	576.8990	173986.2174	6/16/2025
## 2	2	Low	0.9990260	611.6961	-3791.2684	6/16/2025
## 3	3	High	0.9999101	570.3925	138747.3478	6/12/2025
## 4	4	Low	0.9979098	621.5734	3224.2535	6/12/2025
## 5	5	High	0.9989882	590.7374	-109278.7391	6/18/2025
## 6	6	Low	0.9990995	590.3138	954.7682	6/18/2025



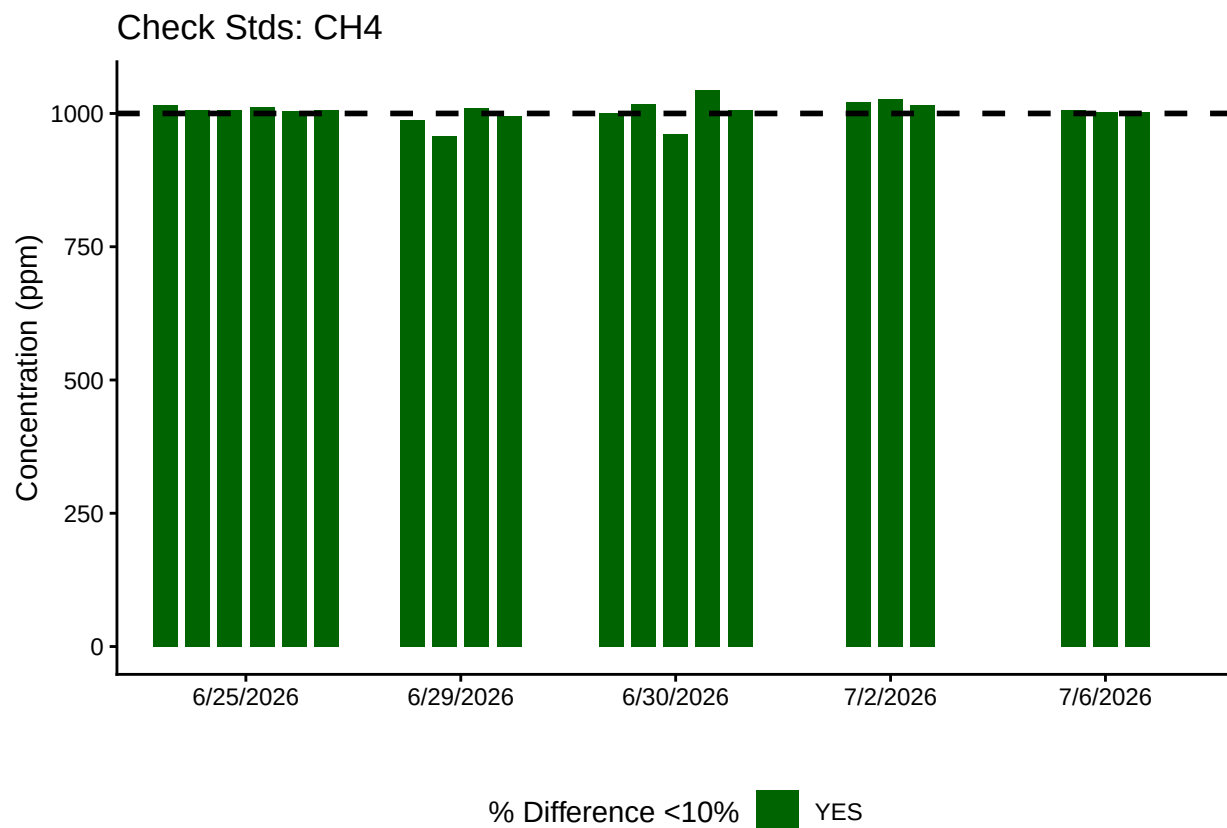
```
## [1] "High_R2 is Good-6/25/2026" "High_R2 is Good-6/29/2026"
## [3] "High_R2 is Good-6/30/2026" "High_R2 is Good-7/2/2026"
## [5] "High_R2 is Good-7/6/2026"
```

```
## [1] "Low_R2 is Good-6/25/2026" "Low_R2 is Good-6/29/2026"
## [3] "Low_R2 is Good-6/30/2026" "Low_R2 is Good-7/2/2026"
## [5] "Low_R2 is Good-7/6/2026"
```

2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

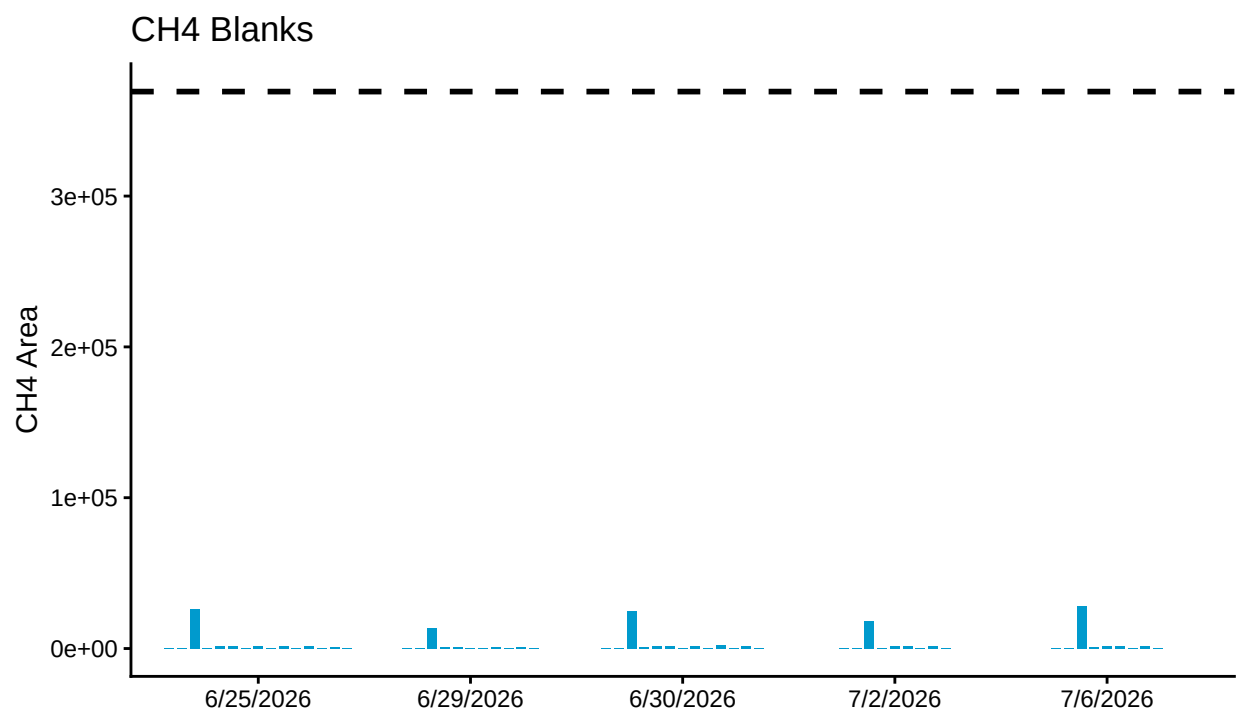
```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

Assess Blanks

>60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

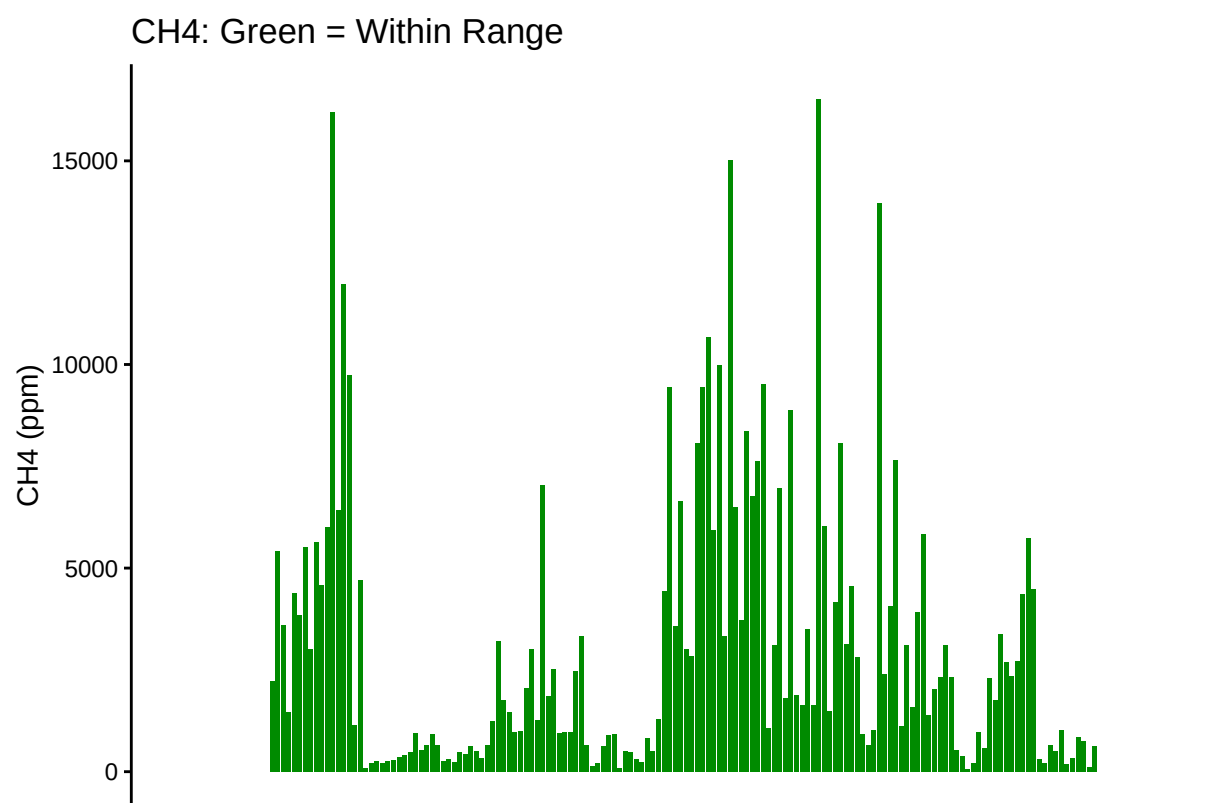
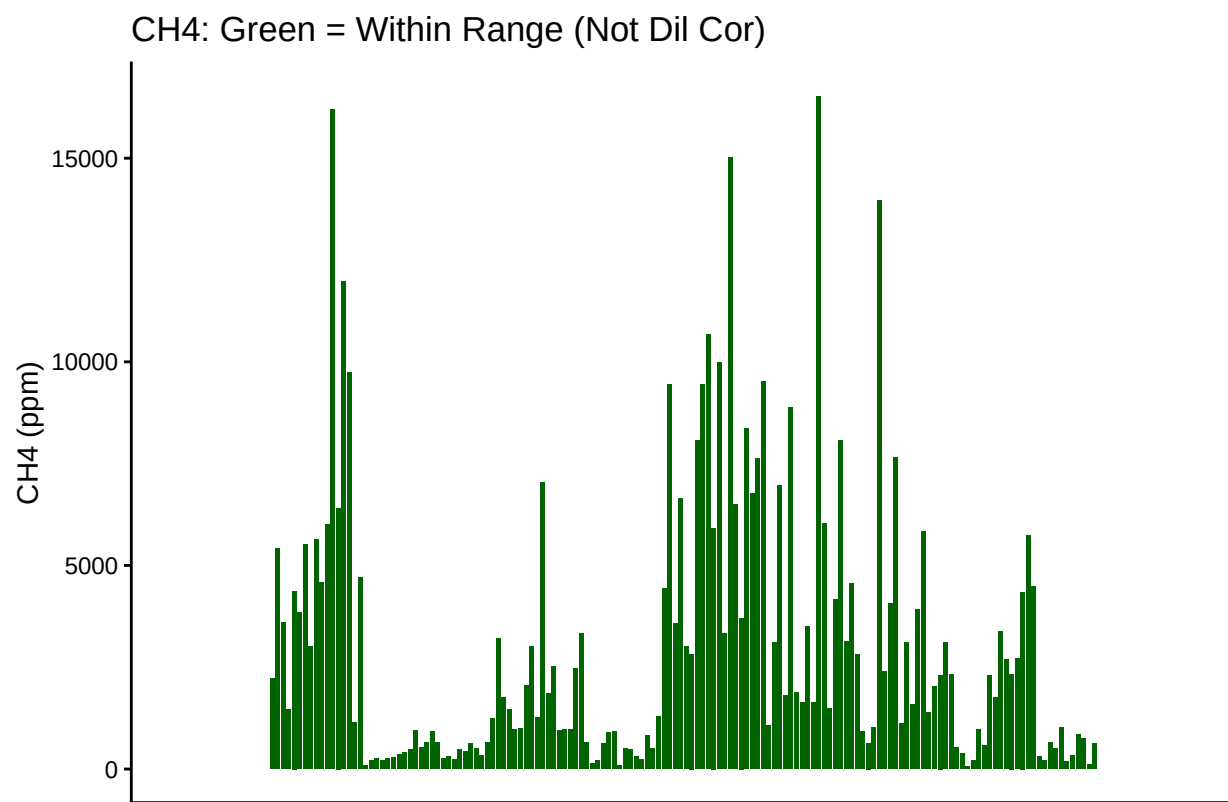


Blank Conc <25% Quartile Samples ■ YES

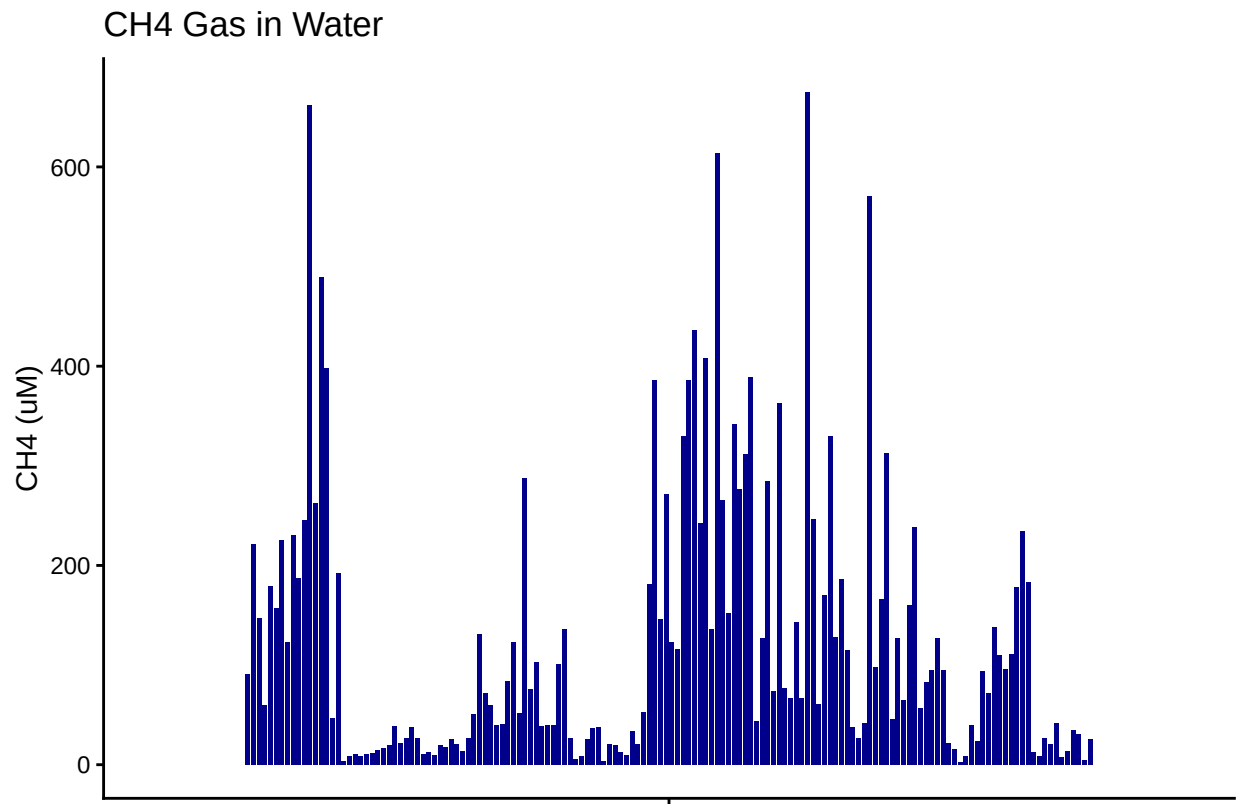
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	2649.737

5 Dilution correct samples and Check Range



6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

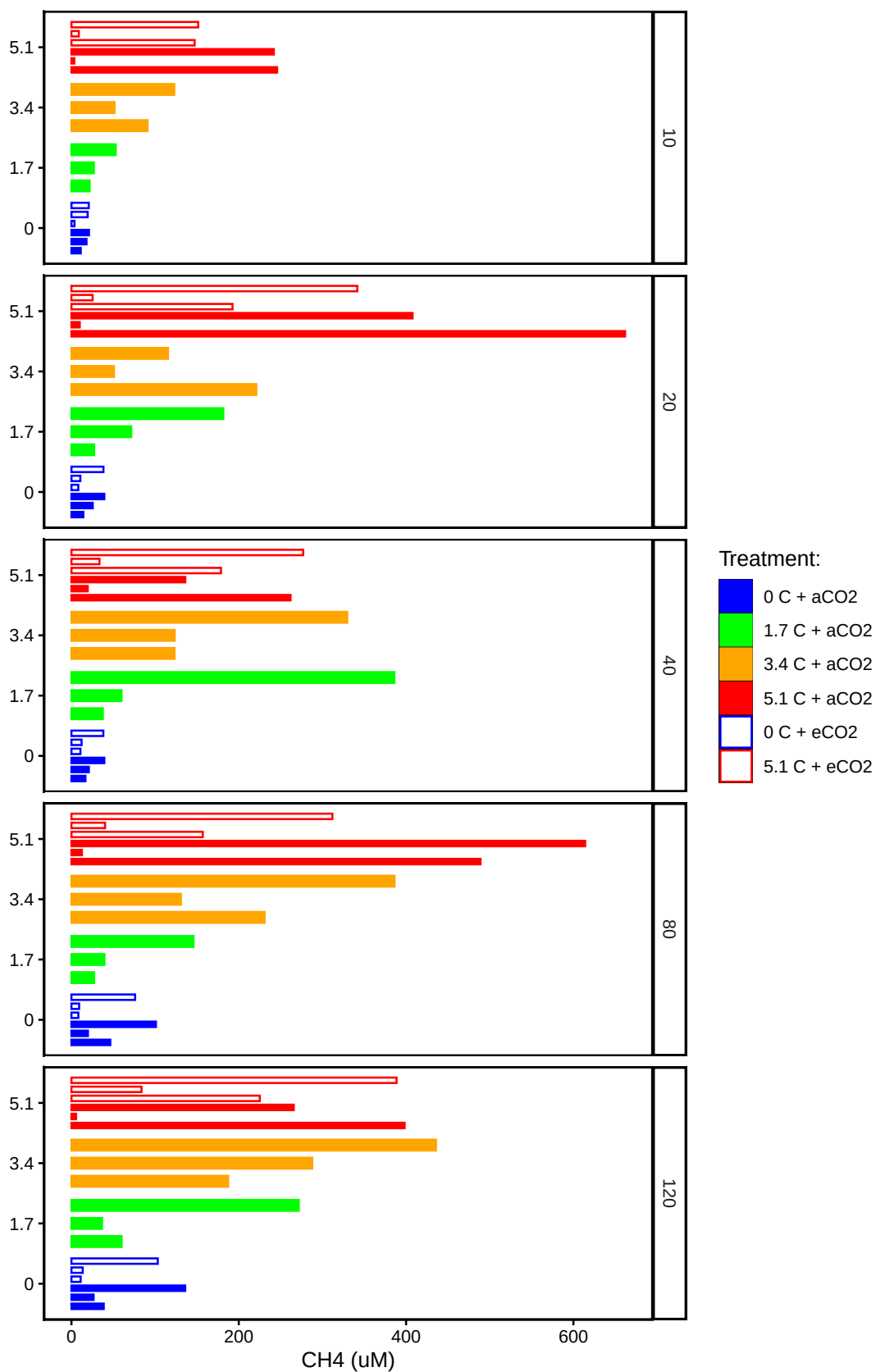
```
## All sample IDs are present in data
```

7.1 Organize Data

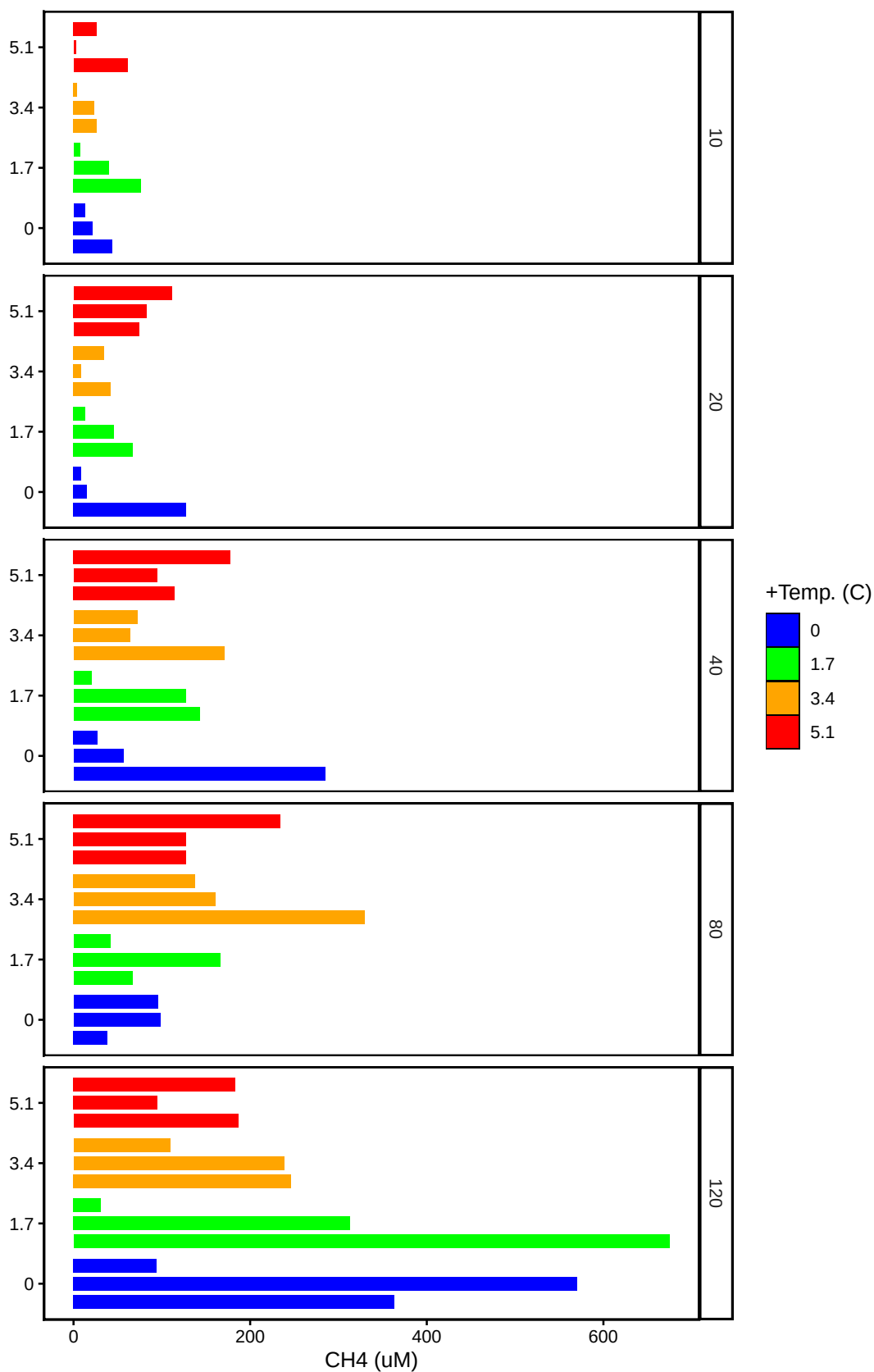
7.2 Visualize Data

```
## Visualize Data
```

C3: Porewater CH4

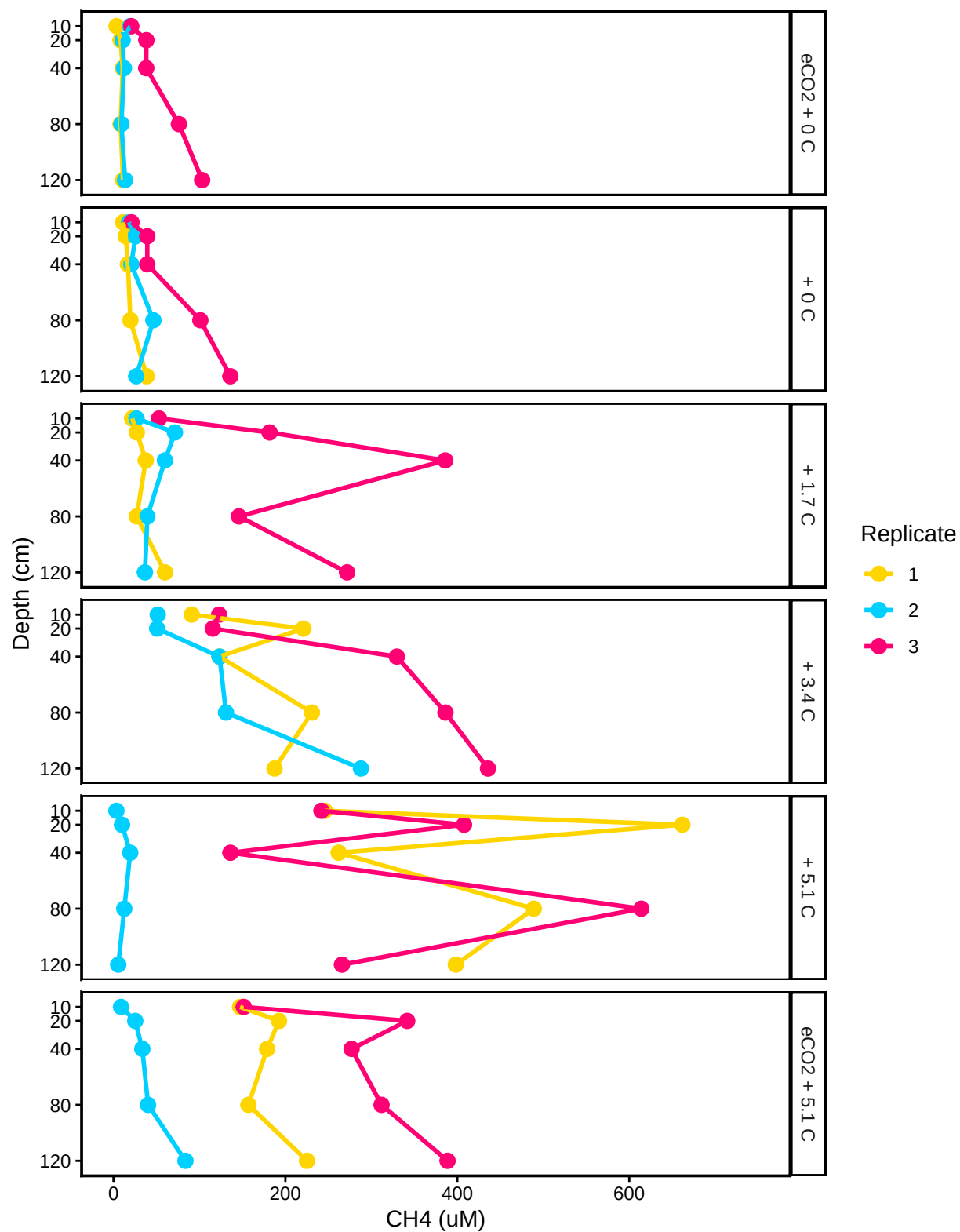


C4: Porewater CH4

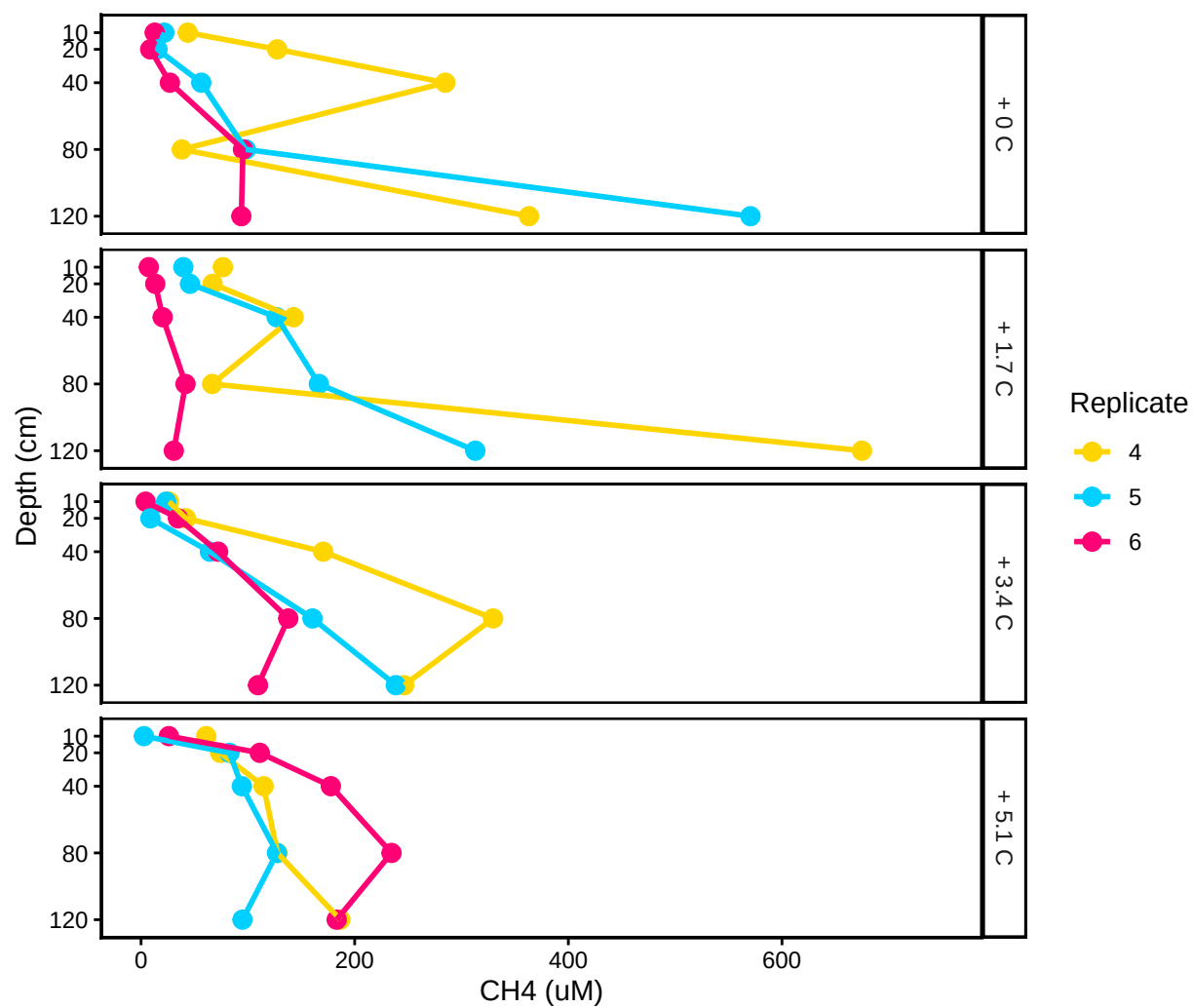


```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use 'linewidth' instead.  
## This warning is displayed once per session.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

C3: Porewater CH4 by Replicate



C4: Porewater CH4 by Replicate



##Export Data

#end